

Doruk Karademirler - RESUME

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EDUCATION

University of Toronto | Honors Bachelor of Science (Expected)

Sep. 2021 - Nov. 2026

Computer Science Specialist, Economics Minor, 3.50 GPA

Mississauga/Toronto, ON

- Participated in the PEY Co-op Program.
- Relevant courses: Algorithm Design and Analysis | Introduction to Artificial Intelligence | Software Tools and Systems Programming | Data Structures and Analysis | Multivariable and Vector Calculus | Probability and Statistics I | Software Design | Introduction to Machine Learning | Introduction to Software Engineering | Introduction to Databases.

PROFESSIONAL STATEMENT / SUMMARY

Computer science student with 5 years of experience in computer-aided design and developing problem-solving algorithms. Proficient in programming languages such as Java, JavaScript, Python, CSS, and C. Skilled in utilizing Git for version control and developing CI/CD pipelines for compiler toolchains. Experienced in using technical expertise to analyze and interpret economic data.

WORK EXPERIENCE

SOFTWARE TOOLCHAINS ENGINEERING INTERN

July 2024- September 2025

Qualcomm

Markham, ON

- Diagnosed and resolved a performance bottleneck in the expf.c function of Android’s math library, eliminating 4K page crossing and improving HDR value consistency. Achieved a 28% performance boost in Geekbench benchmarks.
- Designed and implemented a comprehensive **CI/CD pipeline** using GitLab to automate build, test, and publish stages for GCC and LLVM toolchains. Created and configured **virtual machines** to validate pipeline functionality and benchmark performance using **SPEC CPU 2017**, ensuring consistent and scalable compiler optimization across AArch64 and X86 targets.
- Implemented a pointer chase optimization algorithm for both GCC and LLVM compilers, reducing execution cycles for load instructions that reuse the same register and improving memory access efficiency. This resulted in a 3% performance gain in SPEC benchmarks with pointer-heavy data structures.

TEACHING ASSISTANT

Sept. 2023 - January 2024

UNIVERSITY OF TORONTO

Mississauga, ON

- Conducted 6 individual tutorial sections (40 students per session) and office hours for CSC236 - Introduction to the Theory of Computation, achieving a 15% improvement in student performance on course assessments.
- Responsibly graded assignments and exams, ensuring timely feedback to students while maintaining consistent grading standards.
- Collaborated closely with the course instructor to align teaching strategies with the course objectives, resulting in a cohesive learning experience.

PROJECTS

Price of Empire (Research Opportunity Program) | Visual Studio Code, R, Python

Summer 2023

The research project focuses on creating a comprehensive meta-database accessible through the UTM Economics website, providing researchers with detailed Canadian trade data. Under the guidance of Professor Nicholas Zammit, developed a robust Python algorithm to cleanse and structure a complex economic dataset using the pandas library, which will be used to analyze the British colonial relationships on Dominion countries like Canada. Worked closely with a diverse team of economists and computer scientists to merge domain-specific knowledge with technical approach.

Software Developer - CourseBind Website | ReactJS, CSS, PHP, JavaScript, SQL

January 2024

As a key member of a five-person development team, contributing to the design, development, and deployment of CourseBind, a comprehensive educational platform designed to enhance student productivity and learning. Demonstrated proficiency in full-stack web development, from conceptualization to deployment, within a collaborative, Agile team environment.

Causal Inference with Directed Acyclic Graphs (DAG)

November 2023

Implemented Variable Elimination for exact inference and Likelihood Sampling for approximate inference on Directed Acyclic Graphs (DAGs). Explored causal structures using real COVID-19 data from February and March 2020. Developed Causal Bayesian Networks to assess the causal effects of variables on COVID-19 mortality. Utilized Python and Bayesian network libraries for efficient implementation.